CV

CostVision

Manufacturing Should-Cost Intelligence Platform

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Spur Gear, 38T, m3, 30mm face, Ø40 bore

Commodity: GEAR · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 6 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥121.08	**29.3%**	**36.3%**	**9.3%**	**7.4%**

Model Confidence: **Low** · 6 traced operations · 13 data points auditable

Engine Warnings (3)

88 of 88 gear shop parameters are representative values, not verified data (source set: "representative (no plant data supplied)"). Cutting speeds, feeds, tool life and heat-treat rates were not taken from a standard, a machine-builder catalogue or your plant. This estimate shows the right structure and is NOT a quotable number until the shop data is replaced.

Gear route: Gear hobbing -> Chamfer and deburr -> Induction harden and temper -> Gear grinding -> Gear inspection — cutting cycle 3.60 min/part; perishable tooling £0.59/part; NRE £24000 amortised over 200,000 parts.

Hardening is in the PROCESS bucket, not material: induction harden and temper runs 52 s/part on gear-induction (single-shot coil (Ø120 mm within the 250 mm encircling limit): 2.5 s/module × m3 = 8 s heat + 6 s quench + 20 s in-line temper + 18 s handling). Unlike a furnace bought by the kg, induction hardening is machine time.

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 266.0 cm³; mass for the selected material family 2.088 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: gear

Alloy / material: 4140 / 42CrMo4 / EN19 (Chromoly Alloy Steel) · Net weight: 2.088 kg (measured)

Material utilisation: 100% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 6

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥35.46	29.3%	
2. Process (Machine)	¥44.01	36.3%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
3. Direct Labour	¥11.25	9.3%	
4. Tooling (amortised)	¥6.47	5.3%	
5. Packaging	¥1.18	1.0%	
6. Logistics	¥2.08	1.7%	
Factory Cost	**¥100.45**	**83.0%**	
7. Overhead (SG&A)	¥11.66	9.6%	
Subtotal	**¥112.11**	**92.6%**	
8. Supplier Margin	¥8.97	7.4%	
TOTAL SHOULD-COST	**¥121.08**	**100.0%**	

§2 — Commercial Parameters

Overhead Rate	**12.0%**	Supplier Margin	**8.0%**
Packaging / part	**¥1.18**	Logistics / part	**¥2.08**
Total Tooling Cost	**¥1293885.60**	Amortisation Volume	**200,000 parts**

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	**mat-steel4140**	ID	
Grade / Specification	**4140 / 42CrMo4 / EN19 (Chromoly Alloy Steel)**		UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. | Regional adj. ×0.880 (millSteel)
Region	**China**		
Net Finished Weight	**2.0880 kg**	kg	Weight in finished part
Direct Material Cost	**¥35.46**	CNY	Bypasses weight-based model

Data Confidence	**Medium**		
Effective Date	**2026-07**		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Blank turning (face, OD, bore)	CNC Lathe (2-axis)	¥185.20	4.93	1	80.0%	**¥19.04**
Gear hobbing	CNC Gear Hobber — small ("d m 4 , "d Ø 2 0 0)	¥265.03	1.16	1	80.0%	**¥6.40**
Chamfer and deburr	Gear Chamfer / Deburr Cell	¥123.89	0.43	1	80.0%	**¥1.10**
Induction harden and temper	Gear Induction Hardening Cell (RF + quench + temper)	¥241.47	0.88	1	80.0%	**¥4.42**
Gear grinding	Generating Gear Grinder (threaded wheel, "d m 6)	¥603.72	0.88	1	80.0%	**¥11.09**
Gear inspection	CNC Gear Checker (profile / lead / pitch)	¥272.86	0.34	1	80.0%	**¥1.95**
TOTAL						**¥44.01**

4B Labour Detail

Operation	Labour Grade	Rate / hr	Manning	Lab. min	Eff %	Labour Cost	Op Total
Blank turning (face, OD, bore)	Skilled Machinist	¥71.49	1	4.93	90.0 %	¥6.53	**¥25.57**
Gear hobbing	Skilled Machinist	¥71.49	1	1.16	90.0 %	¥1.53	**¥7.94**
Chamfer and deburr	Semi-skilled Operator	¥49.77	1	0.43	90.0 %	¥0.39	**¥1.50**
Induction harden and temper	Skilled Machinist	¥71.49	1	0.88	90.0 %	¥1.16	**¥5.58**
Gear grinding	Skilled Machinist	¥71.49	1	0.88	90.0 %	¥1.17	**¥12.26**
Gear inspection	CMM / Quality Inspector	¥72.40	1	0.34	90.0 %	¥0.46	**¥2.41**
TOTAL						**¥11.25**	**¥55.26**

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø40.0 × 30 thru	1	Helical mill / bore	2.00	¥8.48	Yes
Face	8865 mm²	2	Face milling (facing)	2.62	¥11.09	Yes
TOTAL costed		**3**		**4.62**	**¥19.56**	**Yes**

§5 — Machine Rate Buildup

CNC Lathe (2-axis) [mach-lathe-cnc] · Confidence: Low · UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥199095.02	¥62.22	4,000 hr/yr available
Maintenance	¥109502.26	¥34.22	
Energy	¥55085.58	¥17.21	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥109502.26	¥34.22	
Finance Cost	¥79638.01	¥24.89	
TOTAL — CNC Lathe (2-axis)	¥592642.14	¥185.20	Effective hours: 3200/yr

CNC Gear Hobber — small ("dm4, "dØ200) [gear-hob-small] · Confidence: Low · REPRESENTATIV
~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥308597.29	¥96.44	4,000 hr/yr available
Maintenance	¥159276.02	¥49.77	
Energy	¥71611.25	¥22.38	
Floor Space	¥54751.13	¥17.11	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC Gear Hobber — small ("dm4, "dØ200)	¥848081.84	¥265.03	Effective hours: 3200/yr

Generating Gear Grinder (threaded wheel, "dm6) [gear-grind-generating] · Confidence: Low · R
(threaded-wheel) gear grinder class. Target ~£130/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @78% util	Notes
Depreciation	¥821266.97	¥263.23	4,000 hr/yr available
Maintenance	¥338461.54	¥108.48	
Energy	¥151485.34	¥48.55	
Floor Space	¥109502.26	¥35.10	
Indirect Support	¥288687.78	¥92.53	
Finance Cost	¥174208.14	¥55.84	

TOTAL — Generating Gear Grinder (threaded wheel, "dm6)	¥1883612.04	¥603.72	Effective hours: 3120/yr

Gear Chamfer / Deburr Cell [gear-deburr] · Confidence: Low · REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @82% util	Notes
Depreciation	¥129411.76	¥39.45	4,000 hr/yr available
Maintenance	¥69683.26	¥21.24	
Energy	¥33051.35	¥10.08	
Floor Space	¥39819.00	¥12.14	
Indirect Support	¥89592.76	¥27.31	
Finance Cost	¥44796.38	¥13.66	
TOTAL — Gear Chamfer / Deburr Cell	¥406354.52	¥123.89	Effective hours: 3280/yr

Gear Induction Hardening Cell (RF + quench + temper) [gear-induction] · Confidence: Low · REPRESENTATIVE — UK gear induction hardening cell. Energy-heavy (RF generator + quench recirculation). Target ~£55/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥308597.29	¥96.44	4,000 hr/yr available
Maintenance	¥129411.76	¥40.44	
Energy	¥115679.72	¥36.15	
Floor Space	¥44796.38	¥14.00	
Indirect Support	¥99547.51	¥31.11	
Finance Cost	¥74660.63	¥23.33	
TOTAL — Gear Induction Hardening Cell (RF + quench + temper)	¥772693.29	¥241.47	Effective hours: 3200/yr

CNC Gear Checker (profile / lead / pitch) [gear-checker] · Confidence: Low · REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @65% util	Notes
Depreciation	¥288687.78	¥111.03	4,000 hr/yr available
Maintenance	¥109502.26	¥42.12	
Energy	¥27542.79	¥10.59	
Floor Space	¥59728.51	¥22.97	
Indirect Support	¥149321.27	¥57.43	

Finance Cost	¥74660.63	¥28.72	
TOTAL — CNC Gear Checker (profile / lead / pitch)	¥709443.24	¥272.86	Effective hours: 2600/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.
Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	3.9184	£	Pre-computed by commodity module	mat-steel4140	Medium
Blank turning (face, OD, bore).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Blank turning (face, OD, bore).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear hobbing.machineRatePerHr	29.2853	£/hr	REPRESENTATIVE — UK small CNC hobber class. Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-hob-small	Low
Gear hobbing.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Chamfer and deburr.machineRatePerHr	13.6897	£/hr	REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-deburr	Low
Chamfer and deburr.labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low
Induction harden and temper.machineRatePerHr	26.6821	£/hr	REPRESENTATIVE — UK gear induction hardening cell. Energy-heavy (RF generator + quench recirculation). Target ~£55/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-induction	Low
Induction harden and temper.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear grinding.machineRatePerHr	66.7113	£/hr	REPRESENTATIVE — UK generating (threaded-wheel) gear grinder class. Target ~£130/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-grind-generating	Low
Gear grinding.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear inspection.machineRatePerHr	30.1513	£/hr	REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-checker	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
Gear inspection.labourRatePerHr	8.0000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-inspector	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥121.08 ± 16.6%	¥101.63	¥120.00	¥141.90	moderate	13.1%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 12 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
Import duty and international freight (regional table is Ex-Works).
Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Direct material cost	¥35.46	-3.5%	+3.5%	**¥8.58**
Blank turning (face, OD, bore): Cycle Time	0.08hr	-2.6%	+2.6%	**¥6.19**
Blank turning (face, OD, bore): Machine Rate	¥185.20/hr	-1.9%	+1.9%	**¥4.61**
Gear grinding: Cycle Time	0.01hr	-1.2%	+1.2%	**¥2.97**
Gear grinding: Machine Rate	¥603.72/hr	-1.1%	+1.1%	**¥2.68**
Overhead %	12%	-1%	+1%	**¥2.52**
Blank turning (face, OD, bore): Labour Rate	¥71.49/hr	-1%	+1%	**¥2.52**
Gear hobbing: Labour Rate	¥71.49/hr	-1%	+1%	**¥2.52**

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥40.30	¥80.01	¥40.51	¥11.76	¥15.55	**¥188.14**	¥1.44	**¥200.22**	+65%
Germany (EUR)	¥41.50	¥84.02	¥65.47	¥12.35	¥20.15	**¥223.50**	¥1.65	**¥237.56**	+96%
France (EUR)	¥41.10	¥73.61	¥47.06	¥10.82	¥16.33	**¥188.93**	¥1.65	**¥201.36**	+66%
Spain (EUR)	¥40.30	¥70.41	¥29.67	¥10.35	¥12.90	**¥163.63**	¥1.65	**¥174.83**	+44%
Poland (PLN)	¥39.09	¥57.61	¥18.41	¥8.47	¥9.46	**¥133.05**	¥1.72	**¥142.67**	+18%
Turkey (TRY)	¥36.27	¥48.01	¥10.23	¥7.06	¥7.14	**¥108.70**	¥1.87	**¥117.16**	-3%
China (CNY)	¥35.46	¥44.01	¥11.25	¥6.47	¥11.66	**¥108.85**	¥2.08	**¥121.08**	Base
India (INR)	¥36.27	¥41.61	¥7.16	¥6.12	¥5.91	**¥97.07**	¥2.15	**¥105.01**	-13%
Mexico (MXN)	¥38.28	¥48.01	¥11.87	¥7.06	¥7.39	**¥112.61**	¥1.94	**¥121.24**	+0%
United States (USD)	¥40.30	¥68.01	¥53.20	¥10.00	¥14.68	**¥186.18**	¥1.79	**¥198.47**	+64%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

8.85 kgCO2e · 4.24 /kg · cradle-to-gate

Material	**4.38 kgCO2e**	Material factor	**2.10 kg/kg (Steel)**
Process	**4.13 kgCO2e**	Grid intensity	**0.550 kg/kWh · 7.52 kWh**
Logistics	**0.33 kgCO2e**	Total	**8.85 kgCO2e**

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

§11 — Cost Intelligence Insights

1 findings · ~8% combined saving

[INFO] Split routing — consolidation worth quoting

Finding	6 machining operation(s) across 6 separate stations. Each station is a re-fixturing: fixture cost, handling time, and datum-transfer variation.
Impact	Medium · up to 8% potential saving
Action 1	Quote a multi-axis (4/5-axis) consolidation of the milled and drilled features against the split routing
Action 2	Redesign features (deep pockets, undercuts) that force additional setups

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥14.77/part (~12%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
1	Process & Automation	**Re-quote Machining on the Cost-Optimal Routing**	The costed 6-station routing books £8.39/part of machining; the same content on the cheapest capable machines costs £7.61 — a £0.79/part (9%) machine-mix saving. Note: 5-axis consolidation was ranked and does NOT win on this part.	**¥11.34**	9.4%	Quick Win	Low
2	Process & Automation	**OEE Improvement Programme (TPM)**	OEE at 80% vs 85% world-class target. Each 5% OEE gain reduces effective machine cost by 5%.	**¥1.82**	1.5%	Medium Term	Low
3	Design & Geometry	**Tolerance and Surface-Finish Relaxation**	Grinding/honing/reaming content is 10.1% of the part cost. Each of those operations exists to hit a callout — challenge the callouts before paying for them.	**¥7.26**	6.0%	Quick Win	Low
4	Design & Geometry	**Design Review for DFM Compliance**	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	**¥6.05**	5.0%	Quick Win	Low
5	Design & Geometry	**Fix the burr at source (tooling discipline, gate redesign); where finishing must remain, tumble/vibratory beats manual**	Manual finishing after the primary process — Deburring/fettling/rework appears as its own operation — a symptom of an upstream condition (tool wear, gating, die state) being paid for by hand, every part.	**¥3.63**	3.0%	Quick Win	Low
6	Commercial & Sourcing	**Annual Volume Re-Commitment for Better Pricing**	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	**¥3.63**	3.0%	Quick Win	Low
7	Sustainability & Energy	**Energy Productivity Programme**	This process is energy-intensive (melt, cure, oven or heat-treat content). Energy sits inside the machine rate and overhead — and it is the fastest-moving input cost of the decade.	**¥3.63**	3.0%	Medium Term	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	2	Re-quote Machining on the Cost-Optimal Routing	**¥11.34**	¥13.16
Design & Geometry	3	Tolerance and Surface-Finish Relaxation	**¥7.26**	¥16.95
Commercial & Sourcing	1	Annual Volume Re-Commitment for Better Pricing	**¥3.63**	¥3.63
Sustainability & Energy	1	Energy Productivity Programme	**¥3.63**	¥3.63

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Re-quote Machining on the Cost-Optimal Routing
- Tolerance and Surface-Finish Relaxation
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing